

Figure 1: Ganglia architecture

cluster state to a central *gmetad* daemon, which is responsible for data aggregation and presentation.

Ganglia also comes with a unique PHP front-end that enables IT admins to get complete diagnostic and real-time information of the state of your clusters. Since Ganglia stores metric data over a period of time, the IT admin can now view historic data of the clusters and hosts ranging from the past hour, day, week, month, and even year.

Introducing sFlow

While monitoring physical servers is a standard task even on a large scale, monitoring virtual infrastructure is much more challenging. A cloud's virtual compute infrastructure can scale up and down dynamically depending on various factors such as CPU load, disk read writes, network I/Os, etc. With such unpredictability, it becomes really difficult for a monitoring tool to gauge whether a virtual machine is actually experiencing problems or not.

sFlow is an industry standard, network monitoring protocol that offers end-to-end monitoring for numerous devices and applications. It provides a standard set of metrics that can be collected from a wide variety of operating systems, virtualisation platforms, and physical devices such as switches, routers, etc. These standardised metrics are fully compatible with those offered by Ganglia, thus reducing complexity in configuration and management.

Similar to Ganglia's *gmond* daemon, sFlow provides a metric data collector in the form of the Host sFlow agent.

Host sFlow agent

This is an open source implementation of the sFlow standard protocol, and is used for physical and virtual server monitoring. The Host sFlow agents are also available for monitoring hypervisors, including Xen, KVM/Libvirt, Hyper-V and VMware ESX, and are supported on a variety of OSs as well, such as Windows, Linux, Solaris, etc.

The Host sFlow agent provides scalable, multi-vendor, multi-OS performance monitoring with minimal impact on the systems being monitored.

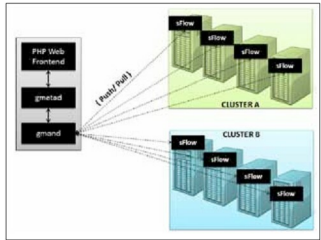


Figure 2: sFlow-Ganglia architecture

Thus, IT admins now have the ability to monitor and gather metric data not only from the physical servers (using Ganglia), but also from the virtual machines and the applications that are running on them, thus providing a complete picture of how the cloud's compute infrastructure is performing.

Setting up Ganglia and sFlow in the cloud

In this guide, we are going to set up a Ganglia and sFlow monitoring system that will be able to dynamically monitor and report the performance of cloud instances in real-time.

These steps were performed on Amazon Web Services EC2 computer servers, but can be replicated identically on any major cloud services provider including Eucalyptus, Openstack, Cloudstack, etc.

Figure 3 gives the set-up diagram. In this scenario, we are using two instances for demo purposes. One instance will contain the Ganglia *gmetad* and *gmond* daemons, and the other instance will contain only the Host sFlow agent. The Host sFlow agent will continuously monitor the instance and send back the collected metrics to Ganglia's *gmond* daemon, which in turn, will feed the metrics to *gmetad* for processing and visualisation.

In a full-fledged production environment on the cloud, there can be multiple such Host sFlow agents monitoring the CPU, network, disk workloads and reporting these back to a single or even multiple instances of Ganglia. The design can vary as per your requirement.



Note: In this guide, I have compiled Ganglia from source as the sFlow integration capabilities are not yet supported on the Ganglia RPM binaries.

Installing and configuring Ganglia

The first thing to do is to install some pre-requisite RPMs:

```
# yum install gcc gcc-c++ autoconf automake expat-devel
libconfuse-devel rrdtool rrdtool-devel apr-devel libconfuse
```

Next, download and compile the latest version of PCRE (Perl Compatible Regular Expressions)